

Reproducing OpenWAM-α on LIBERO

Closed-loop success rate on a single A100

Yutao-Yang

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Motivation

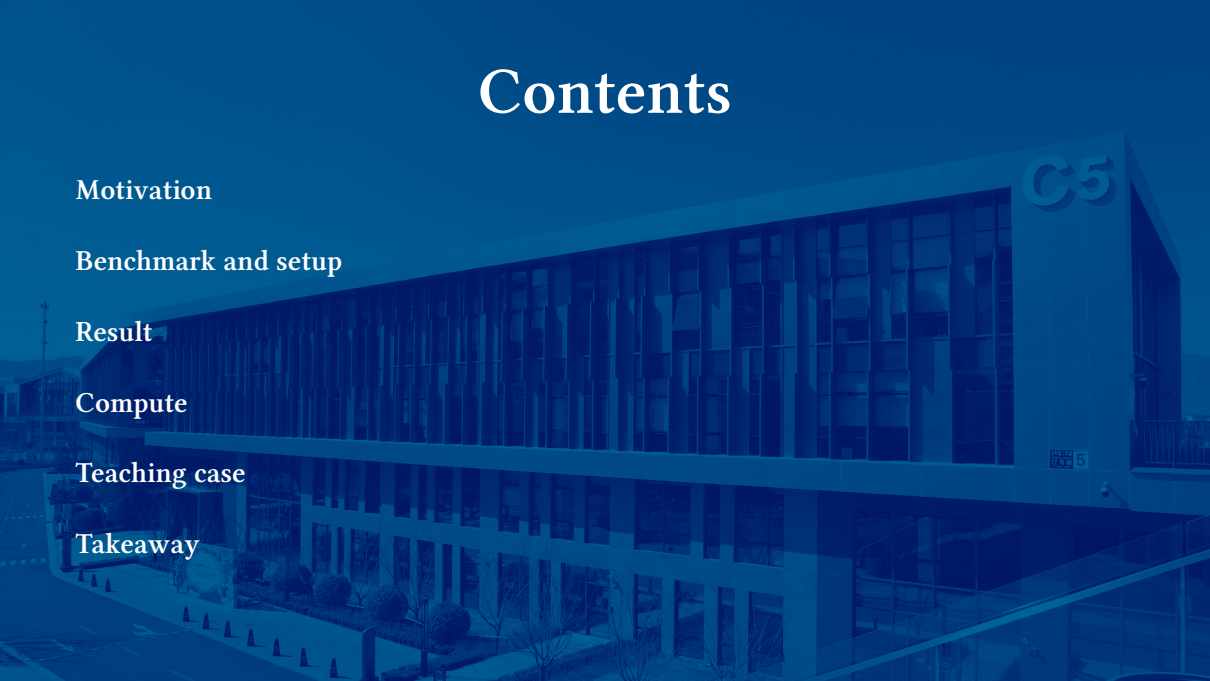
Benchmark and setup

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Motivation

Why reproduce OpenWAM-α?

- OpenWAM-α is a world–action model: a video-generation backbone is joined with an action expert.
- The authors report **99.30 %** on the four standard LIBERO suites.
- Question: does the *released* checkpoint reproduce that number under the *native* task protocol?

What is held fixed

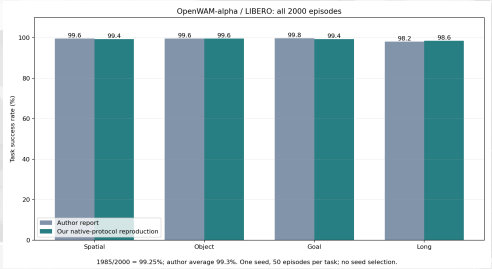
Released weights, LIBERO task protocol, environment and budgets. No retraining, no per-seed tuning.



Benchmark and setup

LIBERO at a glance

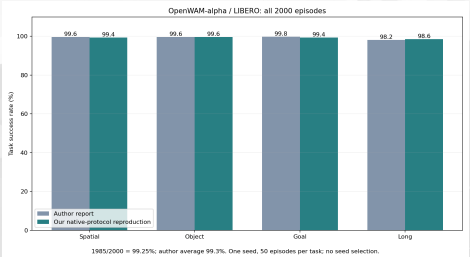
- Four suites: Spatial, Object, Goal, LIBERO-10.
- 40 tasks $\times$ 50 episodes $=$ **2000** closed-loop episodes.
- 20 Hz control; per-task budgets of 600 / 700 steps.
- Success is decided by the task's BDDL goal predicate.





Result

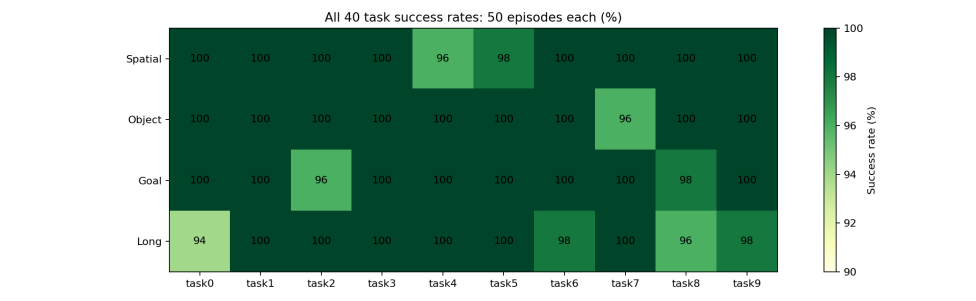
- Reproduced: $\mathbf{1985/2000 = 99.25\,\%}$.
- Official: $99.30\,\%$ (-0.05 percentage points).
- All 2000 episodes retained; engineering errors reported separately.



Per suite

Spatial $497/500$ Object $498/500$ Goal $497/500$ LIBERO-10 $493/500$.

Every task, every suite





Compute

Compute and control budget

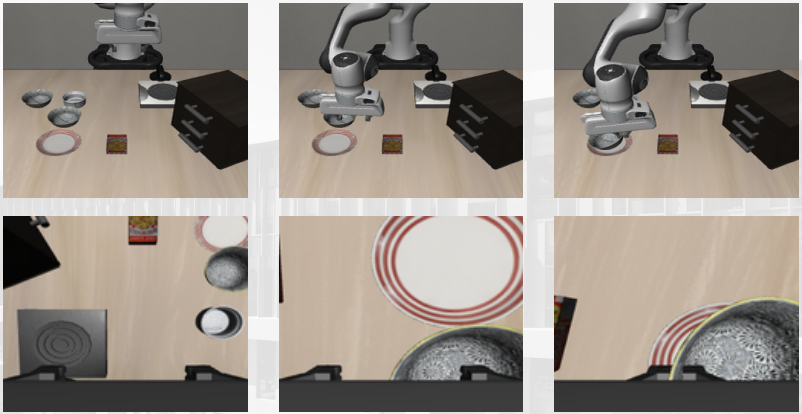
- Wall time: 25 829.198 s.
- Peak memory: 25 676 MiB on one A100.
- Mean GPU utilisation: 29.40 %.
- 301 810 executed control steps.





Teaching case

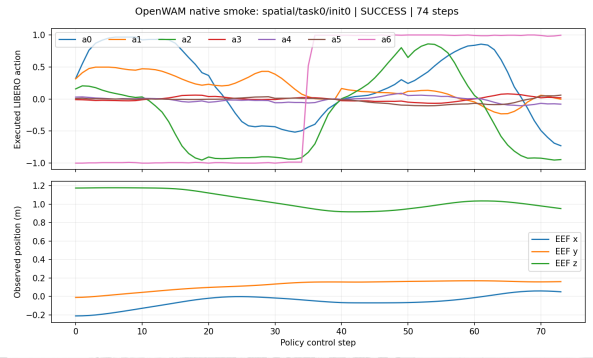
A genuine closed loop: head and wrist views

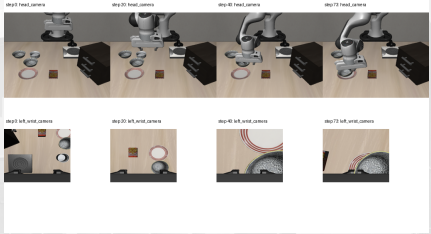


step 0 step 36 step 73

External (top) and wrist (bottom) frames at three control steps.

Action trace and rollout





Play video



Takeaway

What this shows --- and what it does not

- The released checkpoint **reproduces the official score** to within 0.05 percentage points.
- Agreement is about the aggregate number, not per-episode identity.
- No world-model video metric is claimed: the run stored no predicted future frames.
- The result must not be read as a fair architecture comparison with the other model in this project.

- OpenWAM: *An Open, Modular Exploration Towards Systematic World–Action Model Pretraining*, arXiv:2609.07398.
- LIBERO: Liu et al., *Benchmarking Knowledge Transfer for Lifelong Robot Learning*, arXiv:2306.03310.

Full report: `reports/2026.10.05-2026.10.11-openwam-libero/`